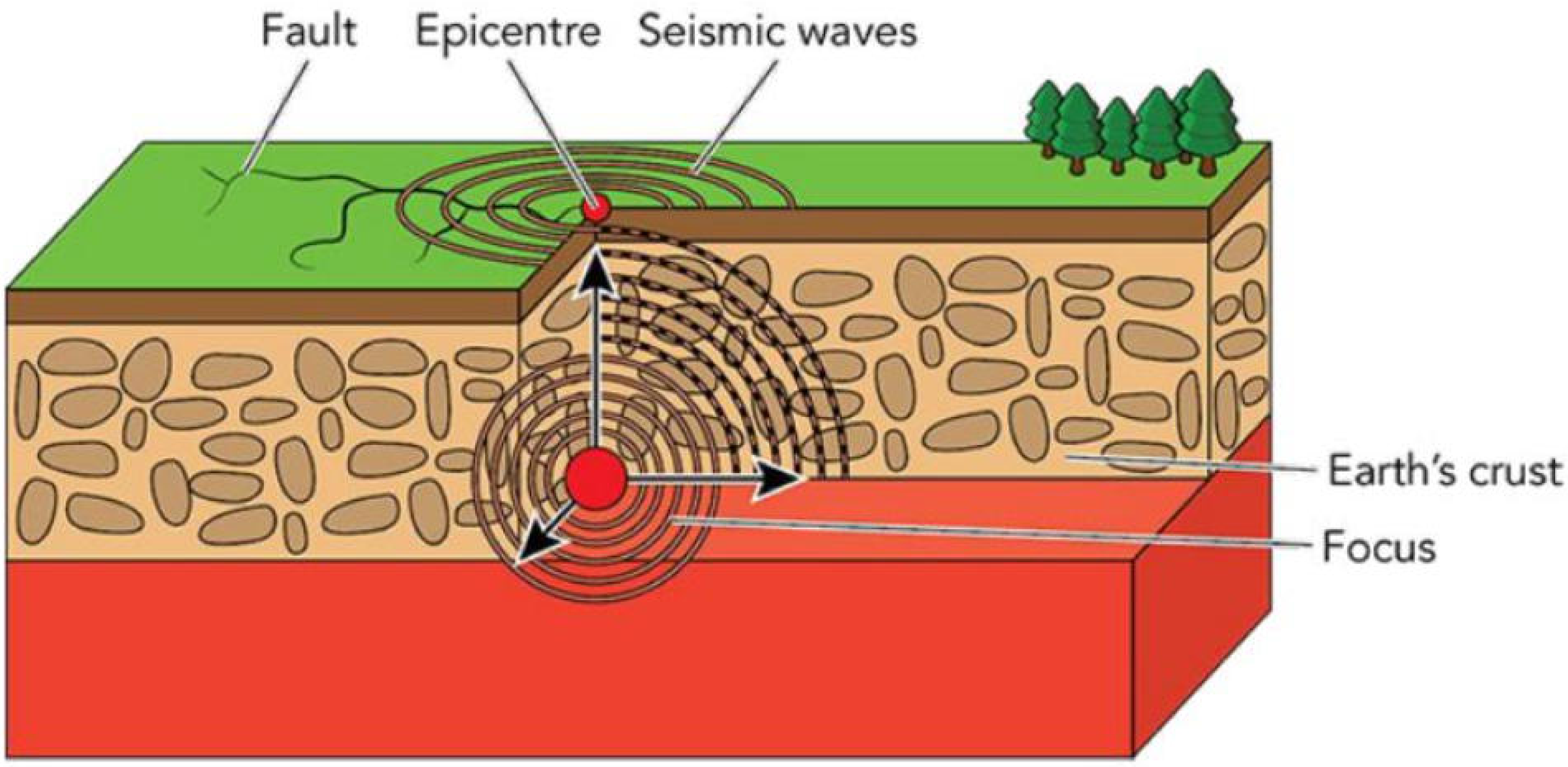
4.2 The processes and features of earthquakes and volcanoes

Understanding the processes underlying an earthquake or volcanic eruption helps you to understand the hazards that they pose and how to manage their impacts.

The processes and characteristics of earthquakes

Earthquakes are caused by the movement of large slabs of rock beneath the Earth's surface that are under great stress and pressure. As the plates move, the rocks can become stuck, building up stress and tension. When sufficient stress has built up, the rocks slip along the fault line which can be hundreds of km long. This results in the release of large amounts of energy.

The point at which the rocks start to move is the **focus** or **origin**. The point directly above the focus, where the energy from the earthquake first reaches the Earth's surface, is the **epicentre**. The moving rocks release energy in shockwaves which spread out from the focus. These shockwaves are called **seismic waves**. It is these waves that cause the shaking and damage observed during an earthquake. Earthquakes occur along all types of plate boundaries.



Source A: The main characteristics of an earthquake

TASK 1: Study Source A

a Explain the difference between the focus and the epicentre of an earthquake.

b Explain why an earthquake would be felt the most strongly at the epicentre.

c In pairs, discuss why an earthquake can be felt at great distances from the epicentre. Write two bullet points supporting your answer.

d Do you think that the depth of the earthquake below the surface would affect the strength of the earthquake at the surface? Discuss this in pairs and write down a brief answer giving your reasons. Then, share your ideas with the class.

Earthquake hazards

There are a wide variety of hazards that earthquakes cause:

1. The shaking of the Earth's surface: this causes the collapse of buildings and infrastructure, and slope failure which can cause harm.
2. Liquefaction: when soil or sand is destabilised, as groundwater is forced to the surface. This causes the ground to take on more liquid characteristics. This can result in the collapse of buildings as their foundations become unstable.
3. Ground displacement: when land moves up and down or sideways, resulting in a visible step or offsetting of the Earth's surface. This can range from a few centimetres to many metres. This fracturing of the Earth can cause damage to buildings and infrastructure.
4. Flooding: an earthquake can result in the failure of dams that then causes flooding. The flooding can cause harm to people and infrastructure.
5. Tsunami: the displacement of the seafloor during an earthquake can trigger a tsunami and cause coastal flooding. Any towns, housing, or infrastructure along the coastline are at risk of being washed away.
6. Slope failure and landslides: when the slopes on a hillside weaken due to the shaking of the Earth. Any material flowing or falling down the hillside can cause harm to people or infrastructure as it moves over the area.

Source B: Visible ground displacement across a road, caused by an earthquake

TASK 2: Study Source B

a Describe the movement of the land that caused this type of damage during an earthquake.

b If the soil had a high water content during this earthquake, what other hazard would you expect to see? How could this hazard present a danger to the people in the affected area?

c In pairs or small teams, research the cause of the tsunami in Japan in the 2011 earthquake. **Create an annotated diagram** to explain how the movement of the Earth resulted in the tsunami forming.

The processes and characteristics of volcanoes

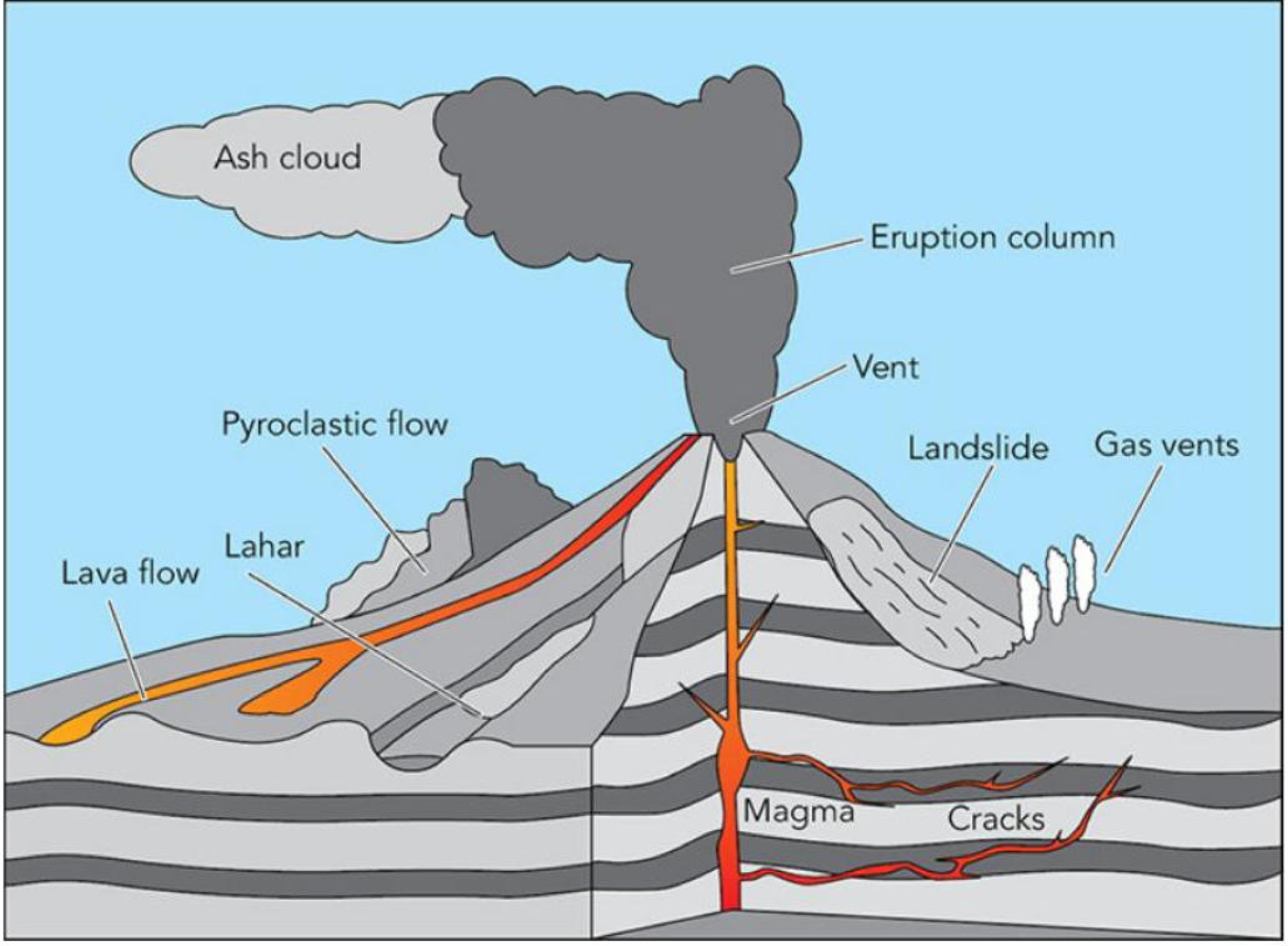
A volcano is a geographical feature in the Earth's crust. Volcanoes occur where molten rock is squeezed between cracks in the Earth's crust at such high pressures that it reaches the surface of the Earth. Volcanoes form in different ways:

1. along converging plate boundaries where subduction occurs and results in the formation of a magma chamber under the surface
2. at diverging plate boundaries where magma rises to the surface where plates are separating.

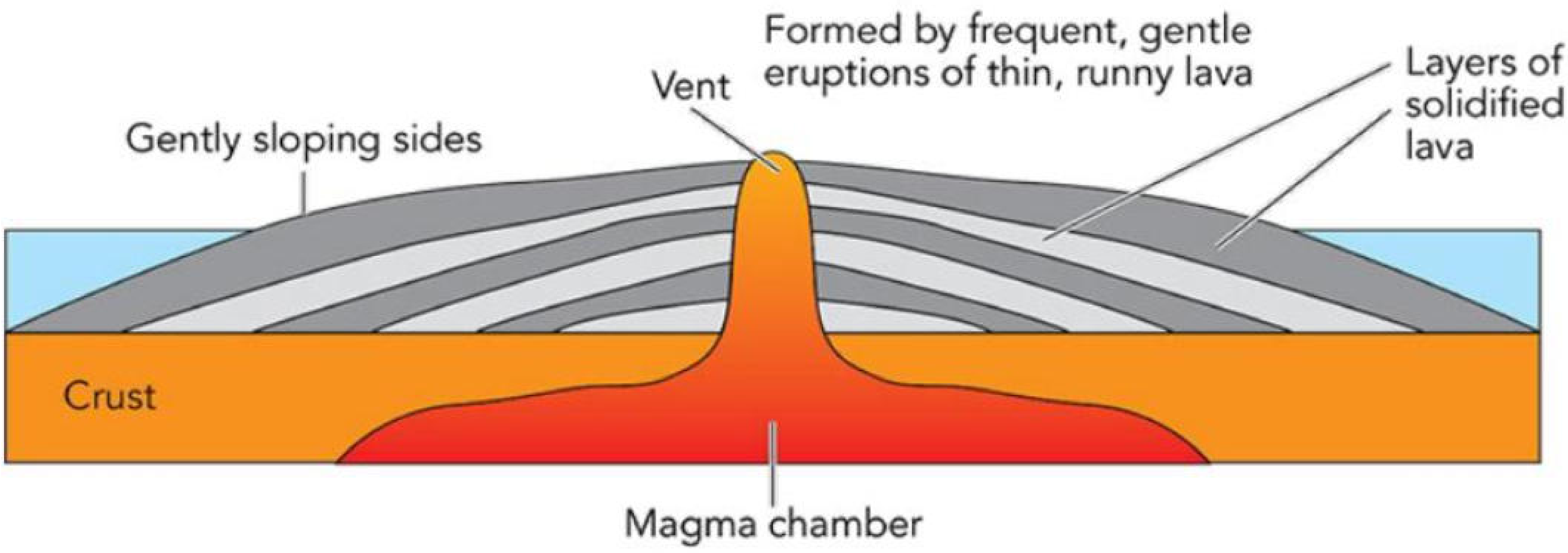
Common characteristics of volcanoes include:

- Magma: molten rock is called magma when it is beneath the surface, and lava when it reaches the surface.
- **Magma chambers:** a large pool of liquid rock beneath a volcano where molten rock collects. The magma is less dense than the surrounding rock which results in it being driven upwards through cracks in the surrounding rock.
- **Volcanic vents:** the point at the Earth's surface where lava flow, rock fragments, and gases erupt.
- **Craters:** at the surface after an eruption has occurred. Craters are bowl or funnel-shaped depressions that usually lie directly above the volcanic vent.
- Secondary vents or cones: volcanoes may have more than one vent. These secondary vents are smaller than the main vent and may form additional craters on the sides of the volcano.

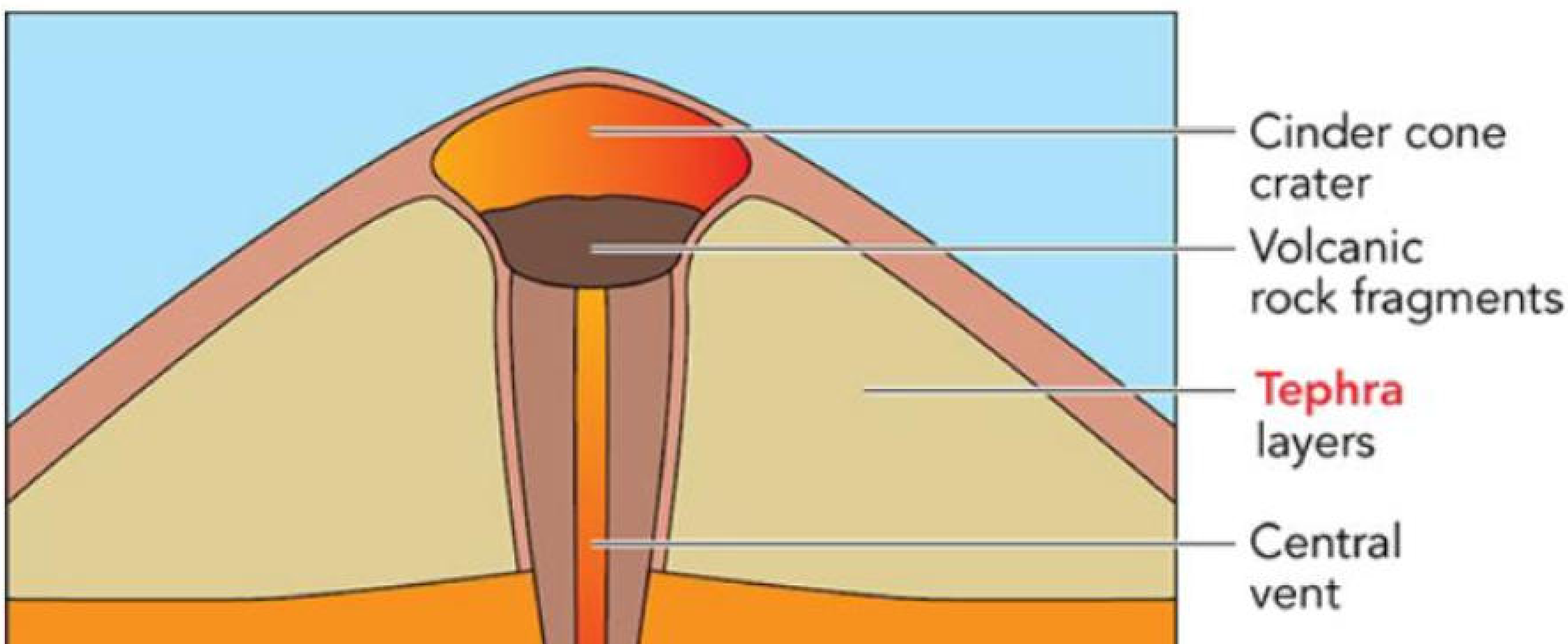
Volcanoes form in a wide variety of shapes and sizes. The three main types of volcanoes are stratovolcanoes (also known as composite volcanoes), shield volcanoes, and cinder cone volcanoes. These volcanoes each have different characteristics and are found on different types of plate boundaries (Table 4.1).



Source C: Stratovolcanoes



Source D: Shield volcanoes



Source E: Cinder cone volcanoes

Table 4.1: Volcano characteristics

Strato/Composite cone	**Shield volcano**	**Cinder cone volcano**

volcano		
Usually found on convergent/destructive plate boundaries.	Usually found on divergent/constructive plate boundaries.	Found on divergent/constructive or convergent/destructive plate boundaries
Areas where magma has high pressure build-up under the Earth's crust in magma chambers.	Magma build-up under the Earth's crust is under less pressure than stratovolcanoes. A weakness in the crust allows magma to 'leak' through.	Smaller eruptions, unlikely to cause deaths. Most cinder cones erupt only once, and eruptions tend to be weak. They often form as secondary cones along the side of larger shield or stratovolcanoes. They form quickly (within a year) when gas forces hot lava upwards.
Lava is very hot and viscous (thick and sticky), which results in explosive eruptions. The lava does not flow easily.	Lava is generally hot and very fluid, flowing easily. The lava can flow great distances from the volcanic vent. Shield volcanoes are only explosive if water gets into the vents, with the water turning to steam and increasing pressure in the volcano. Otherwise, they have low-explosivity.	Gas-charged volcanic material is blown into the air. This breaks into small fragments as it falls to the ground and solidifies, forming a hard cone. These types of volcanoes usually occur in groups. As the vent beneath the Earth moves, new cones form next to each other at the surface.
Steep sided, large (up to 3500 metres high), cone shaped volcano. Can have more than one crater, fed by more than one vent.	The volcano has gentle sloping sides and is shield-like in shape, can reach up to 8500 metres in height.	The most common type of volcano. Cone-shaped with steep slopes and small in size (100–400 metres high). A single bowl-shaped crater fed by a single vent.
Stratovolcanoes are usually about 50% lava and 50% **pyroclastic** material in alternating layers. This gives them their other common name: composite volcanoes.	90% of the volcano is lava rather than pyroclastic material.	The cone is made up of blobs of lava that have hardened to form cinder cones.
Examples: Mount St Helens (USA), Mount Etna (Italy), Mount Pinatubo (Philippines).	Example: Surtsey (Iceland).	Examples: Parícutin (Mexico), Wizard Island (USA) and Sunset Crater (USA).

We can also classify volcanoes by the frequency of their activity. The states they may be in are:

- **Extinct:** there is no evidence of activity for 10 000 years or more, and they are unlikely to erupt again.
- **Dormant:** these volcanoes have not erupted in a long period of time but are expected to erupt again in the future due to a magma chamber still being present.
- **Active:** these volcanoes have a recent history of eruptions and are likely to erupt again.

TIP

There are always exceptions to the rule. For example, there are shield volcanoes with steep slopes, and you can find some shield volcanoes on destructive plate boundaries. Another example is that a volcano may seem extinct, having been inactive for over 10 000 years, but it is possible that it erupts again in the future. As with everything in nature, categories are general guidelines.

TASK 3: Study Sources C D E

a Create three spider diagrams to compare the characteristics of shield volcanoes, cinder cone volcanoes, and stratovolcanoes.

b Explain why a shield volcano is less likely to have steep sides than a stratovolcano.

c Which of the three types of volcanoes do you think is more dangerous? Explain your answer.

d **Draw a sketch diagram** of a stratovolcano. Include the labels: magma chamber, vent, crater, secondary vent, and lava.

TASK 4

Referring to Source B in Topic 4.1 and the information in Table 4.1, write 100 to 200 words to explain the distribution of volcanoes that you observe on the map. Include the types of plate boundaries and the types of volcanoes you would expect to find.

Volcanic hazards

Like earthquakes, volcanoes have hazards. These can cause harm and have impacts on humans, the economy, and the environment.

Millions of people live on or near active volcanoes. This number is increasing with population growth. As a result, more people are at risk from the hazards that volcanoes present. Figure 4.1 describes some common volcanic hazards.

Figure 4.1: Volcanic hazards

Earthquakes: volcanic eruptions can trigger earthquakes

Lava flows: steams of molten rock that pour from an erupting vent.

Pyroclastic flows: a hot (typically>800°C) mixture of rock fragments, ash, and gas that travels at high speeds away from the volcanic vent during an eruption.

Lahar: a hot or cold mixture of water, ash, and rock fragments that flow rapidly down the slopes of a volcano and extend up to 80 km from the volcano. Their rapid speed of movement means that they can be more deadly than lava flows.

Volcanic hazards

Toxic gases: volcanoes can release significant amounts of carbon dioxide (CO_2), sulphur dioxide (SO_2), and other gases. Depending on their concentrations, these gases can pose a danger to humans, animals, and agriculture.

Ash-fall. During eruptions the ash will:

- fall to the ground around the volcano, landing on buildings and infrastructure
- cover trees and plants and strip them of leaves
- be carried by winds over large distances. The fine ash is made of sharp fragments.

Tephra: materials that are deposited after an eruption by falling through the air. For example, ash particles that fall from a high eruption cloud to form widespread layers downwind from the eruption.

TIP

Tephra and pyroclastic flows are slightly different. The material in a pyroclastic flow is deposited by settling out of the flow close to the volcano. Tephra is deposited by falling through the air and can be found great distances from the volcano.